3. It shows how the linguistic, annotation and evaluation challenges compound one another rather than merely co-occurring, and sets out a research agenda that follows from that interaction.

The literature examined here was identified through searches of the ACL Anthology, IEEE Xplore, Scopus and arXiv, combining terms for the language variety, the sentiment task and annotation quality. Studies that only benchmark classifiers, with no annotation or reliability component, were not included.

The rest of this paper is organised as follows. Section 2 explains why Malaysian code-mixed writing is difficult to annotate. Section 3 examines annotation practice and its failure modes, and Section 4 examines the models that are used as annotators. Section 5 discusses how annotation reliability is evaluated. Section 6 brings these findings together and sets out directions for future research, and Section 7 concludes the review.

2. Why Malaysian Code-Mixed Text Is Difficult to Annotate

Section 1 identified the sentiment labels, rather than the model, as the greater constraint on sentiment analysis of Malaysian Malay-English text. Before annotation practice is examined, the text being annotated needs to be described, because three of its linguistic properties affect the annotation task directly. The first is that switching between Malay and English occurs inside the sentence, so the boundary between the two languages does not indicate where the opinion is located. The second is that the affect of a message, meaning the emotional attitude it expresses, can be carried by Malay particles that appear in no standard sentiment lexicon. The third is that phonetic spelling gives a single word many written forms. All three properties are described in the sociolinguistic literature on Malay-English contact (Muysken, 2013; Treffers-Daller et al., 2022; AlAfnan et al., 2025). Each property is presented below together with its consequence for the annotator, whether human or automatic, who must assign a sentiment label.

2.1 Switching occurs within the sentence and offers little reliable structural cue

Treffers-Daller et al. (2022) analysed 1,044 switched utterances from the classroom speech of two Malaysian teachers of English and found that most of the switches occurred within the clause rather than between clauses. Insertion, in which a word or phrase from one language is placed inside a stretch of the other, accounted for 33.0% of the switches. Congruent lexicalisation, in which words from both languages are combined within a single shared word order, accounted for a further 30.7%. Alternation between whole clauses accounted for 24.8%. The authors attribute this distribution to the structure of the two languages rather than to the style of the speakers. English and Malay share the same basic word order, in which the subject comes before the verb and the verb before the object. Malay also has very little inflectional morphology, meaning that its words change little in form to mark grammatical features such as tense. A speaker can therefore move from one language to the other in the middle of a clause with little grammatical adjustment (Treffers-Daller et al., 2022; Muysken, 2013).

These proportions were measured on spoken classroom data, so they should not be taken as the proportions in social media text. The explanation, however, concerns the structure of the two languages rather than the medium in which they are used, so the same type of switching can be expected in writing. Mohd Shamsuddin et al. (2024) give a written example in the post "Does institut kanser offer free medication cheap chemotherapy". Here the Malay noun phrase *institut kanser*, in which the noun comes before its modifier as Malay word order requires, is placed inside an English question.

The consequence for annotation is that the annotator cannot use the boundary between the two languages to locate the part of the message that carries the sentiment. In monolingual annotation, locating that part is a separate step from judging its polarity. One example is aspect-based sentiment analysis, in which a sentiment label is assigned to each entity or attribute mentioned in a text rather than to the text as a whole (Nazir et al., 2022). A span is a marked stretch of text. In annotation schemes for that task, the annotator first marks the words that express the opinion as a span and then assigns a polarity label to that span (Pontiki et al., 2016). In Malay-English mixed text the annotator has no language boundary to guide that first step, so deciding which span to judge already requires interpretation. Two annotators may therefore assign different labels not because they judge the sentiment differently but because they are judging different parts of the message. Disagreement about which part of a message to judge is therefore separate from disagreement about its polarity, and it arises first.

2.2 Affective meaning is carried by particles that belong to neither standard language

Locating the part of the message that carries the opinion is the first difficulty. A second difficulty arises once that part has been located, because the word that carries the affect may be a Malay particle attached to a clause that is otherwise entirely English. Muysken (2013) terms this construction back-flagging. Treffers-Daller et al. (2022) define it as the attachment of a discourse marker or a coordinating conjunction from a speaker's first language to a stretch of speech in the second language. In their data, back-flagging accounts for 121 of the 1,044 switched utterances, or 11.6%. The particle involved most frequently is *lah*, which the same study reports can attach to a clause that is entirely English, entirely Malay or mixed. Two examples the study reproduces from earlier descriptive work are "Buy this lah" and "you understand lah kan".

The contribution of these particles is to the tone of a sentence rather than to its factual content. AlAfnan et al. (2025) classify particles as a lexical feature of Manglish that adds emphasis or tone and signals politeness or informality. They also list particles among the features that distinguish Manglish from standard English and standard Malay. The same study identifies the softening of instructions as one function of these particles, giving "Don't disturb la, he's busy" as an example in which the particle makes the instruction less direct. Treffers-Daller et al. (2022) describe the same effect for *lah*, which they report can shift an utterance from a highly assertive tone to a polite request or a friendly encouragement. In both accounts the particle leaves what the sentence states unchanged and alters the

attitude the sentence conveys, and that attitude is the property a sentiment label is intended to record.

Two limits on this evidence should be noted. Treffers-Daller et al. (2022) supply frequencies, but their corpus is fifteen hours of recorded classroom teaching by each of two university English lecturers. Their frequencies should therefore not be read as frequencies for social media writing. AlAfnan et al. (2025) describe Malaysian usage more broadly, but their data are scripted dialogue from Malaysian television series such as *Kopitiam* and *KL Gangster*, examined through qualitative content analysis. That study therefore reports what the particles do, not how often they occur. Neither source establishes how frequently particles are used in Malaysian social media posts, and no such figure was found among the studies reviewed here. What both sources do establish is that the construction recurs and that it carries affect.

The consequence for annotation is that the particle indicates the tone of a message but does not determine its polarity. A reader of "Don't disturb la, he's busy" can see that the particle softens the instruction. A softened instruction is nonetheless compatible with a negative reading, in which the writer is irritated, and with a neutral reading, in which the writer is being polite. Treffers-Daller et al. (2022) note the same difficulty for discourse markers as a whole, observing that their specific meanings are often difficult to capture even where their pragmatic functions can be evaluated. The particle therefore carries affect without indicating which polarity that affect has, and two annotators may read it differently while both reading it competently.

Disagreement of this kind is not necessarily reduced by adding annotators. Pavlick and Kwiatkowski (2019) find that where an item admits more than one defensible reading, disagreement between annotators is reproducible and persists as further judgements are collected, rather than converging on a single label. A corpus built by majority vote over such items therefore records the majority reading as though it were the only defensible one, and the variation it conceals is systematic rather than random. An agreement coefficient computed over such items reflects genuine differences in interpretation rather than annotator error alone.

The first property examined in this section produces disagreement about which part of a message to judge. The second produces disagreement about how to judge that part once it has been found, because more than one reading of it is defensible.

2.3 Phonetic spelling multiplies the surface forms of a single word

The first two properties concern where the sentiment in a message is located and how it should be read. The third concerns whether the words that carry it can be recognised at all. Maskat and Rahman (2020) document Malay social media users spelling words as they are pronounced rather than as a dictionary lists them, so that a single word can appear in many written forms. Working from a corpus of 6,241 Twitter posts collected between 2014 and 2018, they sort what they call Social Media Malay Language into five categories: spelling variants, Malay-English mixed sentences, loan words spelled according to Malay pronunciation, slang-based spellings, and vowel-less words.

Each category contains many forms. Among the spelling variants that Maskat and Rahman (2020) record, *macam mana* appears as *mcmana*, *camne*, *camner*, *mcmner* and *cane*. In the vowel-less category a word is reduced to its consonants or to a single character, so *jangan* is written *jgn* and *pada* is written *pd*. The slang category varies by region, with *besar* written *besaq* in the north and *beso* on the east coast. The categories can also combine, and the same study gives *x suko* as a form that belongs to the slang and vowel-less categories at once.

The question that matters for annotation is whether this variation can be reversed automatically, and Maskat and Rahman (2020) tested this directly. They attempted to normalise two of the five categories, spelling variants and vowel-less words, meaning that they tried to map each non-standard form automatically to its standard spelling. The methods used were two measures of string similarity, n-gram similarity and Levenshtein distance, each of which scores how closely two spellings resemble one another. The benchmark was a set of 1,176 pairs of similar words covering 435 distinct words, identified by hand by a native Malay speaker. Precision here is the proportion of the matches proposed by a method that were correct. On that benchmark, the highest precision reported for any configuration was 0.43, and most of the configurations reported were lower than that. The authors conclude that similarity matching can detect the two categories but that a more sophisticated technique is needed, and they state that normalisation of the remaining three categories is left for future research.

Two limits on this evidence should be noted. Only slang from Selangor was handled, so the slang category was tested more narrowly than the regional range the same paper describes. More importantly for this review, English words were removed before the experiment and code-mixed posts were excluded from the evaluation subsets. The reported precision is therefore a measurement on Malay text with the English removed, not on Malay-English mixed text, and no equivalent measurement for mixed text was found among the studies reviewed here.

The consequence for annotation is that human and automatic annotators differ in what they can recognise, and the difference can be seen in the study just described. A single native speaker identified 1,176 word pairs by hand, reliably enough for the result to serve as the ground truth. Automatic matching on the same ground truth reached a highest precision of 0.43. A human reader recognises *x* as *tidak*; the automatic procedure largely did not. The two types of annotator therefore make errors on different inputs and at different rates.

This difference changes what the choice of annotator means. If human and automatic annotation differed only in cost and speed, the choice between them would be a matter of budget. Because they make errors on different inputs, the choice also determines which errors a corpus will contain, and a corpus labelled entirely by one type of annotator carries that type's systematic errors throughout.

The three properties examined in this section therefore create three separate difficulties for annotation. The first makes the part of the message that carries the opinion difficult to locate. The second leaves the reading of that part open once it has been located. The third makes the words themselves difficult for an automatic annotator to recognise, while a human annotator recognises them readily. Section 3

examines the annotation strategies that have been used in response to these difficulties, and the trade-off each strategy accepts.

3. Annotation Practice and Its Failure Modes

Section 2 described what makes Malaysian code-mixed text hard to label. This section examines what has been done about it. The strategies reported in the reviewed studies are set out first, arranged by the trade-off each one accepts. The distribution of labels they produce is examined next, and the practice of pooling several annotators is examined last, because it is the remedy most often proposed, and the one the regional evidence tests most directly (Artstein and Poesio, 2008; Uma et al., 2021).

3.1 Annotation strategies and the fidelity-scalability trade-off

For low-resource code-mixed text, label fidelity and scale are in tension. A corpus can be labelled carefully or it can be labelled in volume. The strategies reported by the studies reviewed here are best read not as competing solutions to one problem, but as different decisions about which of the two to sacrifice. On that reading, the strategies fall into three tiers: manual annotation, semi-automated annotation, and fully automated generative annotation.

The first tier is manual annotation by people. Two of the four Malay-English sentiment resources were built this way, and they are the only two whose labels come from human judgement. Kasmuri and Basiron (2019) drew 5,849 sentences from 23 personal blogs, selected 4,191 for annotation, and retained 2,316 in the finished corpus, reporting $\kappa = 0.83$ for the judgement of whether a sentence was opinionated or factual. Kong et al. (2023) collected 108,246 tweets and manually annotated 11,568 of them into three sentiment classes, using three independent annotators working from a written guideline. What both illustrate is the cost of the decision rather than a defect in the method. Kasmuri and Basiron kept 40% of the sentences they started with, and Kong et al. labelled 11% of what they collected. The labour that produces reliable labels is the same labour that prevents either corpus from reaching the scale a general-purpose Manglish resource would require.

The second tier is semi-automated annotation, which trades in the opposite direction by using existing tools to label at volume and accepting the tools' limitations as the price. Mohd Shamsuddin et al. (2024) built MESocSentiment this way. They collected 229,566 tweets carrying the "#Malaysia" keyword, reduced the set to 138,646 in pre-processing, applied the Malaya and Lingua libraries to identify code-switched posts, and arrived at 20,226 tweets. Sentiment labels were then assigned by majority vote among three existing analysers. The authors state the motivation plainly. The language identification libraries "reduced the need to find fluent speakers in Malay and English to classify the tweets". The reasoning is defensible, and there is evidence from other labelling tasks that automated annotators can match or exceed human crowd workers (Gilardi et al., 2023). What the tier concedes is that the labels can be no more competent than the tools producing them, and inherit whatever those tools cannot read.

The third tier replaces those tools with generative models used directly as annotators. Because a generative model is not restricted to a fixed vocabulary, it can